

No. of Functional Features Included in the Solution

Date	04 July 2026
Team ID	XXXXXX
Project Name	Opticrop-Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

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S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Authentication	Allows users to register, login and manage their profiles securely.	Authentication Module	Done	10%
2	Crop Recommendation	Recommends suitable crops based on soil, weather and location.	Recommendation Module	Done	20%
3	Weather Forecast	Provides real-time weather updates and 5-day forecast for the user location.	Weather Module	Done	15%
4	Soil Analysis	Analyzes soil nutrients and suggests necessary actions.	Soil Analysis Module	In Progress	15%
5	Fertilizer Recommendation	Suggests appropriate fertilizers based on crop and soil conditions.	Fertilizer Module	In Progress	10%
6	Pest Disease Detection	Detects plant diseases from images and provides treatments.	Pest Detection Module	In Progress	10%
7	Market Price Information	Displays current market prices of crops and commodities.	Market Info Module	Pending	10%
8	User Dashboard	Provides an overview of user data, alerts and recommendations.	Dashboard Module	Done	10%

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6

Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	User Registration & Login, User Dashboard, Responsive Design
2	Backend / Logic	3	Crop Recommendation, Soil Analysis, Fertilizer Recommendation
3	Database / Storage	2	User Data Storage, Soil & Crop Data Storage
4	API / Integration	2	Weather API Integration, Market Price API Integration
5	Security / Authentication	1	User Authentication & Authorization